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Building the backend of Radiology

3 Way Partnership

aws

UNIVERSITY of MARYLAND MEDICAL CENTER

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Overview: Medical Images Lifecycle

High frequency — Within organization — Low frequency

Legacy — Amongst organizations — Frontier

Image archive

Acquired images are stored and archived for 7 years.

Current solution:
- PACS
- VNA
- Cloud PACS
- On-premise data center

Problems:
- Lack of compression = storage space
- Security vulnerability
- Costs related to on-premise data center
- Scalability

Analysis by radiologist

Images are analyzed by radiologists at their workstations

Current solution:
- PACS Viewer
- Enterprise Viewer

Problems:
- Latency in loading DICOM images
- Incorrect metadata

Share / Teleradiology

Images are sent to other healthcare systems or to patients

Current solution:
- DICOMweb
- DICOM router
- CD/DVD

Problems:
- Security issues
- Compatability with legacy system e.g. CD, DVD, USB
- Transmission speed

Analysis by AI application

Radiologists use AI applications to analyze medical images

Current solution:
- Integrated case-by-case basis
- Cloud/on-premise computing

Problems:
- Difficult to deploy & integrate AI app
- Each algorithm has narrow use cases

Imaging research

Researchers curate and anonymize datasets from archive

Current solution:
- Download images from in-hospital workstations
- Transport using harddrive

Problems:
- Cumbersome De-ID due to inconsistent meta data
- Large volume

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Petabytes of data, expensive and under-utilized

Large Amount of Data Storage

Imaging data account for 80% of hospital data

Security Vulnerability

Growing threat of ransomware attacks

Long retention period

Regulations require retention of 7 years

Hospitals spend millions of dollars on maintaining on-premise data storage

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Reimagining the Status Quo with Cloud Technology

Current workflow
Doing what has been possible, better

Future-proof
Enabling what hasn't been possible

AWS DICOM API

- Enhanced Security
- Lossless compression
- Faster transmission over the network

Enable Cloud Storage

- More cost-effective & scalable
- Outsource data maintenance

- Web-enabled image sharing
  Image sharing over web to mobile devices
- Open AI application platform
  ML model deployment using cloud computing
- Dataset curation for research
  Imaging data analytics (querying the archive), automated anonymization

... and many more applications

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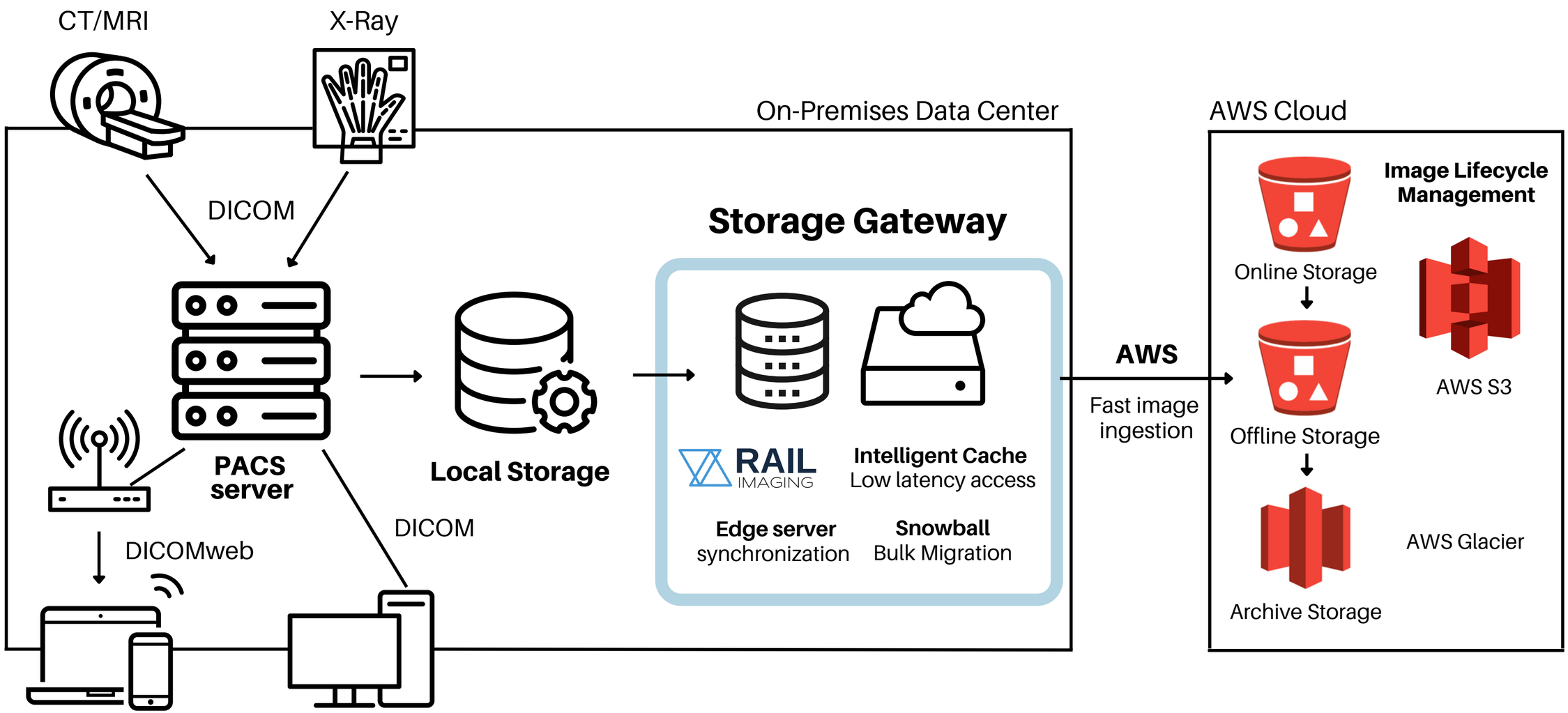
Solution: Cloud Image Management System



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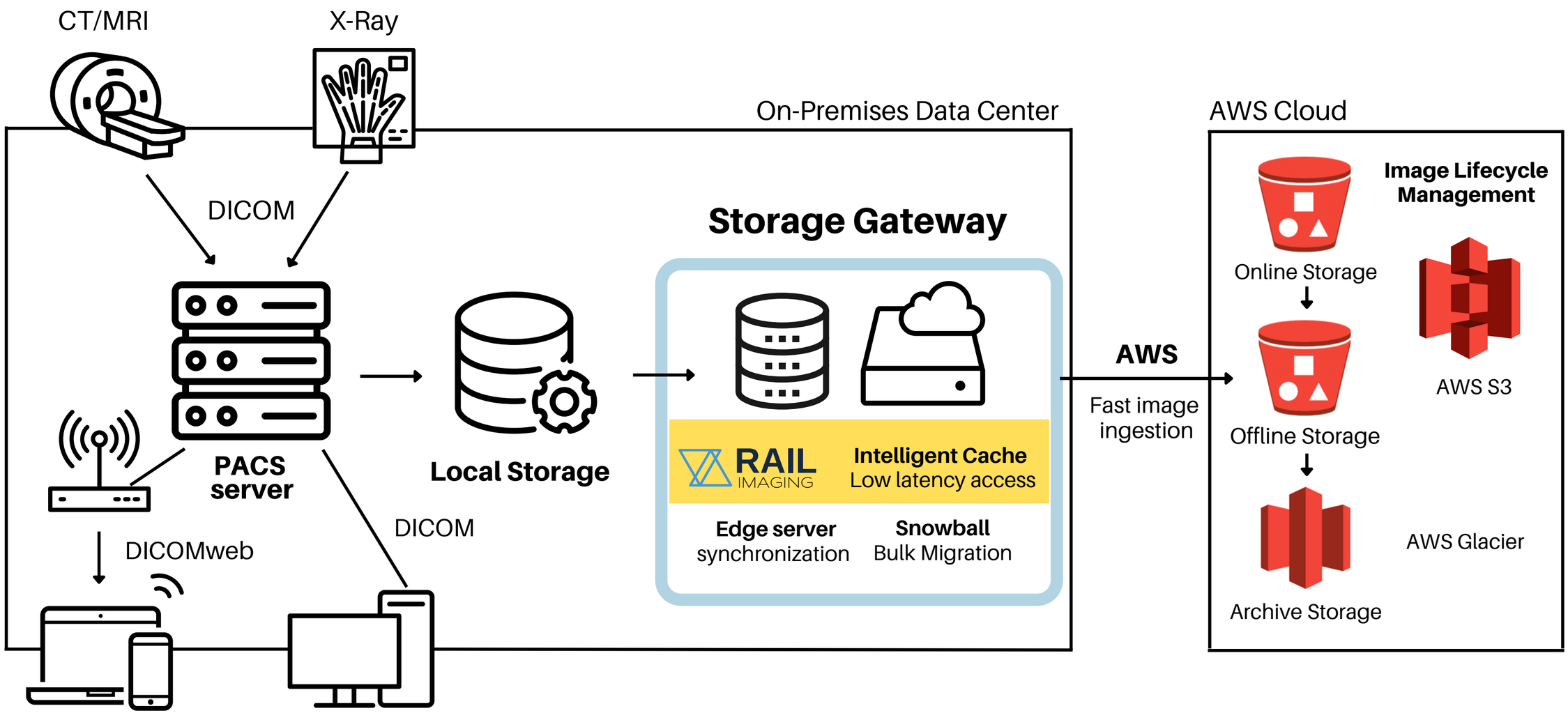

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What is a Storage Gateway?

Storage Gateway is a hybrid cloud storage service that provides **on-premises applications access to virtually unlimited cloud storage.**

Its goal is to deliver low-latency data access to on-premises applications while leveraging the agility, economics and security capabilities of AWS.

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Intelligent Cache
Low latency access through cache

Edge server
On premises & Cloud Synchronization

File Transfer
Secure & Fast DICOM transfer

Snowball
Bulk Migration from PACS to Cloud

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Project Timeline

June - August

- Define the requirement & scope of the pilot study
- Sort out organizational structure and partnerships

August - November

- Build a stand-alone demo based on AWS DICOM API
- Document & measure performance improvements

November 27th: RSNA

- Launch landing page Present results at RSNA
- Collect feedbacks
- Connect with potential customers

December

- Leverage tractions and pilot study to raise seed funding ($1~2M)
- Recruit talents & grow team

2023

- Proof of Concept study with gradient and/or Leidos
- AWS Official Launch

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